

Sample Variance

In investments, we often do not know the mean of a population of interest, so we estimate it using the mean from a sample drawn from the population. The corresponding measure of dispersion is the sample variance or standard deviation.

Sample Variance Formula. The **sample variance**, s^2 , is:

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}, \quad (4)$$

where $\bar{X}$ is the sample mean and n is the number of observations in the sample.

The variance calculation takes care of the problem of negative deviations from the mean canceling out positive deviations by squaring those deviations.

For the sample variance, by dividing by the sample size minus 1 (or $n - 1$) rather than n , we improve the statistical properties of the sample variance. The quantity $n - 1$ is also known as the number of degrees of freedom in estimating the population variance. To estimate the population variance with s^2 , we must first calculate the sample mean, which itself is an estimated parameter. Therefore, once we have computed the sample mean, there are only $n - 1$ independent pieces of information from the sample; that is, if you know the sample mean and $n - 1$ of the observations, you could calculate the missing sample observation.

Sample Standard Deviation

Variance is measured in squared units associated with the mean, and we need a way to return to those original units. Standard deviation, the square root of the variance, solves this problem and is more easily interpreted than the variance.

A useful property of the sample standard deviation is that, unlike sample variance, it is expressed in the same unit as the data itself. If the dataset is percentage of daily returns for an index, then both the average and the standard deviation of the dataset is in percentage terms, while the variance is in squared percentage of daily returns.

Sample Standard Deviation Formula. The **sample standard deviation**, s , is:

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}, \quad (5)$$

where $\bar{X}$ is the sample mean and n is the number of observations in the sample.

Because the standard deviation is a measure of dispersion about the arithmetic mean, we usually present the arithmetic mean and standard deviation together when summarizing data. When we are dealing with data that represent a time series of percentage changes, presenting the geometric mean (i.e., representing the compound rate of growth) is also very helpful.

Downside Deviation and Coefficient of Variation

An asset's variance or standard deviation of returns is often interpreted as a measure of the asset's risk. Variance and standard deviation of returns take account of returns above and below the mean, or upside and downside risks, respectively. However, investors are typically concerned only with **downside risk**—for example, returns below the mean or below some specified minimum target return. As a result, analysts have developed measures of downside risk.